

AI VERTICAL BUSINESS

Zero-Budget Skill & Technical Stack Roadmap

What to learn, in what order, where to learn it for free, and how to find paying customers in a specific industry while you're still learning — written for someone starting with \$0 and a laptop.

Compiled July 2026 · Sources verified against current free-tier terms, course catalogs, and 2026 vertical-AI market data

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01 The 2026 Landscape — Read This Before You Write Code

Since 2023, thousands of founders built the same thing: a chat interface wrapped around someone else's model API, charging a monthly fee. Most of them are gone. Once OpenAI, Google, or Anthropic ship the same feature natively, a thin wrapper's reason to exist disappears overnight — investors and buyers now openly avoid funding "wrapper" products with no owned workflow or data.

What's working instead is **vertical AI**: software that goes deep into one industry's specific workflow (legal discovery, dental scheduling, construction bidding, insurance claims, accounting firm compliance) and owns the messy, industry-specific parts a general model can't — the data, the integrations, the trust, the domain judgment. Vertical AI products are reported to sell at a meaningful premium over general-purpose SaaS, precisely because that depth is hard to copy.

KEY TAKEAWAY

Your unfair advantage was never going to be the AI model — everyone has access to the same three or four frontier models. Your unfair advantage is picking ONE narrow industry, learning its workflow better than a generalist ever will, and building the boring integration and trust layer around the model. That's also the part a zero budget doesn't stop you from doing.

Practically, this means two things for how you should spend your time. First, don't try to master "AI" in the abstract — master a stack that lets you ship a working product fast. Second, start talking to people in a specific industry *before* you're "done learning." Section 05 shows you how to do that at zero cost, in parallel with the technical roadmap below.

02 The Skill Stack — Six Phases, In Order

Learn in this order. Each phase unlocks the next — trying to jump straight to "AI agents" without basic programming and web fundamentals is the single most common reason people stall out.

Phase 0 — Foundations

1–2 weeks

Command line basics, Git & GitHub, Python fundamentals (variables, functions, loops, data structures, virtual environments, pip). This is the floor everything else stands on.

Phase 1 — Web & App Development

4–8 weeks

HTML/CSS/JS, then React and Next.js for the frontend; FastAPI (Python) or Node/Express for the backend; SQL and PostgreSQL for data; REST APIs and basic auth (JWT/OAuth). This is what turns an AI script into a product someone can actually use in a browser.

Phase 2 — AI / LLM Engineering

6–10 weeks, ongoing

Prompt engineering; calling LLM APIs (Claude, Gemini, GPT) with tool/function calling and structured outputs; embeddings and semantic search; Retrieval-Augmented Generation (RAG); vector databases; agent frameworks (LangChain, LlamaIndex, CrewAI, LangGraph); the Model Context Protocol (MCP) for connecting agents to real tools and data. This is the layer that makes your product distinctly "AI."

Phase 3 — No-Code Automation Layer

1–2 weeks, parallel

n8n, self-hosted for free. Lets you wire up a working AI workflow for a real customer in days instead of weeks, before you've committed to a custom codebase. Many of your future customers use Zapier or Make themselves — understanding that world helps you sell into it.

Phase 4 — Deployment & Infrastructure

2–3 weeks

Git-based deployment, environment variables and secrets, frontend hosting (Vercel), backend hosting (Render/Railway), managed Postgres/auth/storage (Supabase), basic Docker. Enough to put a real URL in front of a real customer.

Phase 5 — Business & Distribution Skills

*Start Day 1, ongoing
forever*

Customer discovery interviews, cold outreach, positioning and pricing for a specific industry, Stripe for payments, and "build in public" as a free distribution channel. Most technical founders under-invest here — don't be most technical founders.

03 The Detailed Tech Stack

Everything below is either free and open-source, or has a free tier realistic enough to build and validate a working product on. Paid upgrades only become worth considering after you have a paying customer.

Foundations

Tool	Purpose	Free-tier reality	Learn it at
Git + GitHub	Version control, portfolio, collaboration	Free, unlimited public/private repos	git-scm.com/docs · Learn Git Branching (interactive, free)
Python 3	Backend logic, AI glue code, scripting	Free, open-source	docs.python.org tutorial · CS50x (Harvard, free to audit)
VS Code	Code editor	Free	code.visualstudio.com docs

Web & App Development

Tool	Purpose	Free-tier reality	Learn it at
HTML / CSS / JS	Core of every web interface	Free	The Odin Project (free, full curriculum)
React / Next.js	Customer-facing app UI — the default choice for AI-native startups in 2026	Free, open-source	nextjs.org/learn (free official tutorial)
FastAPI (Python)	API layer connecting your UI to your AI logic; used by ~30% of AI-focused startups	Free, open-source	fastapi.tiangolo.com/tutorial (excellent free docs)
PostgreSQL	Structured data storage	Free, open-source; hosted via Supabase free tier	Full Stack Open (University of Helsinki, free)

AI / LLM Engineering

Tool	Purpose	Free-tier reality	Learn it at
Claude API (Anthropic)	Reasoning, generation, agentic tool use	No permanent free API tier; small one-time signup credit. Prototype prompts free at claude.ai first.	Anthropic Academy — free, 80+ lessons, certificate
Gemini API (Google AI Studio)	Reasoning, generation, multimodal (text/image/audio/video)	Most generous ongoing free tier of any frontier model — no card required, no expiry	ai.google.dev docs
Ollama	Run open models locally	Free, zero API cost — needs a reasonably capable machine	ollama.com docs
LangChain / LlamaIndex / CrewAI / LangGraph	Orchestrate multi-step AI workflows and agents	Free, open-source	DeepLearning.AI short courses (free) · Hugging Face Agents Course (free, certificate)

Tool	Purpose	Free-tier reality	Learn it at
Chroma / Pinecone	Vector database for embeddings (RAG)	Chroma: free, runs locally. Pinecone: free starter index.	Vendor docs + Hugging Face LLM Course (free)
Model Context Protocol (MCP)	Standard way to connect agents to tools and live data	Free, open specification	Anthropic Academy MCP course (free)

Automation, Deployment & Business

Tool	Purpose	Free-tier reality	Learn it at
n8n	No-code AI workflow automation — fastest way to a working pilot	Free forever, self-hosted, unlimited workflows	n8n.io docs + free official courses
Vercel	Frontend hosting for Next.js apps	Free Hobby tier, 100GB bandwidth/month, no card for most uses	vercel.com/docs
Render	Backend/API hosting	Free tier available, no card required (cold starts on free tier)	render.com/docs
Supabase	Managed Postgres, auth, storage, edge functions	Free tier: ~500MB DB, 50k monthly active users, 2 projects	supabase.com/docs
Stripe	Accept payments from customers	Free to integrate — only a % fee per transaction, no upfront cost	stripe.com/docs
Figma	Product mockups and design	Free tier for individuals	figma.com resources (free)

04 The Free Learning Map

Specific, verified-free resources mapped to each phase. Pick **one** resource per row — collecting five courses on the same topic is a common way to feel productive while shipping nothing.

Phase	Resource	What it actually gives you
Foundations	CS50x — Harvard's intro to computer science	Free to audit on edX; covers C, Python, SQL, and the reasoning behind how software is built.
	Learn Git Branching	Free, interactive, visual way to actually understand Git instead of memorizing commands.
Web development	The Odin Project	Free, full-stack, project-based. Deliberately hands-off — you learn by building and getting unstuck, which mirrors real work.
	Full Stack Open (Univ. of Helsinki)	Free, deeper React/Node/TypeScript track if you already know basics and want production-grade depth.
	Next.js official tutorial	Free, maintained by the framework's own team — the fastest path to a deployable app.
AI / LLM engineering	Anthropic Academy	Free, ~80 lessons + certificate, including "Building with the Claude API," "Claude Code in Action," and MCP courses.
	Hugging Face LLM Course + Agents Course	Free, hands-on, code-forward; the agents course includes a free certificate and a final project you build against a benchmark.
	DeepLearning.AI short courses	Free to watch, co-built with OpenAI, Anthropic, Google and LangChain — short, project-based, and current.
	Berkeley LLM Agents MOOC	Free, with guest lectures from Anthropic, OpenAI, Google DeepMind, and NVIDIA on how agents are actually built in industry.
Automation	n8n official docs + free courses	Free; several complete zero-to-hero n8n courses exist with certificates and no signup required.
Business & customers	Y Combinator Startup School	Free, self-paced, includes a library of talks on customer discovery, pricing, and fundraising, plus a founder community.
	"The Mom Test" (concepts, free talks/summaries on YouTube)	Not a course, but the single most useful framework for asking people about their problems without leading the witness.

05 Zero-Budget Customer Discovery — Run This in Parallel

Don't wait until you "know enough" to talk to potential customers. Start in week one. In a market where the moat is domain knowledge, not the model, talking to real buyers early is itself part of the technical work — it tells you what to build.

The zero-budget playbook

- **Pick a niche you already have a foothold in.** A past job, a family member's business, a community you're already part of. Domain access is worth more than a bigger, colder market.
- **Find where that niche already talks online** — a subreddit, a LinkedIn niche, a Facebook group, a Discord or WhatsApp community, an industry forum. This costs nothing but time.
- **Do 15–20 unpaid "problem interviews."** Ask about their current workflow and what's annoying about it — not about your idea. People are far more honest about a problem than about whether they'd hypothetically buy your solution.
- **Look for the same complaint 3+ times.** That repetition is the signal a solo founder with no budget for market research actually needs.
- **Build a "concierge MVP" first.** Solve the problem manually — your own labor plus off-the-shelf AI tools — for one or two willing people before writing a line of custom code.
- **Get a real commitment before you build.** "I'd pay for that" is weak signal. "Here's my card" or "send me an invoice" is real signal.
- **Automate what you were doing by hand** using n8n plus a free-tier LLM API once you've proven the manual version works — this is where Phase 2 and Phase 3 of the tech stack pay off.
- **Build in public.** Post your progress and lessons on X, LinkedIn, or a niche subreddit. It's free distribution and it attracts inbound interest from exactly the kind of early adopter who's willing to try something rough.

THE ORDER MATTERS

The order that actually works for a zero-budget founder is: talk to people → solve it by hand for one of them → get a real commitment → then automate. Building the polished app first and looking for customers second is the most common way solo founders burn months for nothing.

06 Does a Zero Budget Block You From Learning This?

Honestly — no, not the learning part. Almost everything in this roadmap is genuinely free, not "free trial" free. Here's the real breakdown.

Genuinely free, no catch

- Every course listed in Section 04 — freeCodeCamp, The Odin Project, CS50, Anthropic Academy, Hugging Face, DeepLearning.AI, YC Startup School.
- Every core framework and language — Python, JavaScript, React, Next.js, FastAPI, LangChain, n8n (self-hosted) — all open-source with no paywall.
- Google's Gemini API free tier, which has no expiry and no credit card requirement — enough to build and run a real small application while you're learning.
- Free hosting for a first version of your product — Vercel's Hobby tier and Render's free tier both work with no card on file.
- All the customer discovery work in Section 05 — it costs time, not money.

Where money can eventually show up (and how to avoid it early)

- **A custom domain name** (roughly \$10–15/year) — delay this until you have a paying customer; a free Vercel/Render subdomain is fine for validation.
- **Paid LLM API usage** once you outgrow free tiers — Anthropic and OpenAI don't have a permanent free API tier, but Gemini's free tier or a locally-run Ollama model can carry you through the entire learning and early-validation phase.
- **Faster hosting** once you have real users — free tiers have real constraints (cold starts, sleep after inactivity), but they're honestly fine for a pilot with a handful of customers.

The real bottleneck usually isn't money

It's **consistent access to a reasonably capable computer and internet connection**. Course fees are not the obstacle here — everything listed is free. If your own laptop is a real constraint, a library, a university computer lab, or a cheap secondhand machine solves more of the problem than any amount of course budget would.

One more thing worth knowing: once you register an actual company, startup credit programs (Google for Startups, Anthropic and OpenAI's startup programs, AWS Activate) can later hand you anywhere from a few thousand to hundreds of thousands of dollars in compute and infrastructure credit. Treat that as a future unlock once you have traction — not something to chase on day one.

BOTTOM LINE

Zero budget will not stop you from becoming technically capable. It will slow down how much AI API traffic you can push before you have paying revenue, and that's a solvable, temporary constraint — not a wall.

07 13-Week Action Plan

Learning and customer discovery run side by side from week one — not sequentially. Adjust the pace to your own hours available; the order matters more than the exact timeline.

Wk	Technical focus	Customer-facing focus
1	Git, command line, Python basics (CS50x / freeCodeCamp)	Talk to 5 people in a niche you already have access to
2	Python continued + HTML/CSS	Talk to 5 more; write down every recurring complaint
3	JavaScript fundamentals	Pick your niche and specific problem based on the pattern
4	React / Next.js official tutorial	Start a manual "concierge" fix for 1–2 willing people
5	FastAPI + first calls to Claude/Gemini API	Refine the manual process; note what could be automated
6	Prompt engineering (Anthropic Academy / DeepLearning.AI)	Get your first real "I'd pay for this" commitment
7	Embeddings & RAG basics	Build a v0.1 workflow in n8n
8	Vector DB + one agent framework (LangChain or CrewAI)	Onboard your first pilot user on the n8n workflow
9	Supabase (auth/data) + Vercel/Render deployment	Wrap v0.1 in a simple web interface
10	Stripe integration	Ship a paid pilot to 1–3 real customers
11	Fix what breaks based on real usage	Start build-in-public posts about what you're learning
12	Harden the product based on feedback	Expand outreach in the niche using the pilot as proof
13	Review: retention, repeat usage, willingness to pay recurring	Decide: go deeper in this niche, or pivot with evidence

08 Pitfalls That Specifically Kill This Kind of Business

- **Building a thin wrapper.** A chat UI over someone else's model with no owned workflow or data is the exact category investors and buyers have been pulling away from in 2026 — the model provider can absorb your feature overnight.
- **Collecting courses instead of shipping.** A certificate is not a business. One finished, even rough, project beats five completed courses on the same topic.
- **Coding for months before talking to a customer.** Validate the problem manually first; build the software second.
- **Trying to learn every agent framework.** Pick one (LangChain, LlamaIndex, or CrewAI), ship something with it, and only reach for a second framework when you hit a real limitation.
- **Over-building infrastructure before you have users.** Free-tier hosting and a self-hosted n8n instance are enough until you have paying customers proving the demand.
- **Under-investing in outreach.** Most technical founders are far more comfortable writing code than sending the tenth cold message. The tenth message is usually what gets the reply.

09 One-Page Cheat Sheet

Layer	Pick
Languages	Python, JavaScript/TypeScript
Frontend	React → Next.js
Backend	FastAPI (Python) or Node/Express
Database	PostgreSQL, via Supabase free tier
LLM APIs	Claude API, Gemini API (best free tier), Ollama for local/offline
AI orchestration	LangChain, LlamaIndex, or CrewAI + Model Context Protocol (MCP)
RAG / vector search	Chroma (local, free) or Pinecone (free starter index)
No-code automation	n8n, self-hosted, free forever
Hosting	Vercel (frontend) + Render (backend)
Payments	Stripe
Design	Figma
Core free courses	CS50x · The Odin Project · Anthropic Academy · Hugging Face LLM & Agents Courses · DeepLearning.AI · YC Startup School
Customer discovery	Problem interviews → concierge MVP → real commitment → then automate

Free tiers, course catalogs, and platform pricing change — the specific numbers here reflect publicly available terms as of July 2026. Re-check official docs (linked resource names above) before relying on exact limits for a production launch.